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Markdown Studio — Feature Demo

Open this file and run **Markdown Studio: Preview** (`Cmd+Shift+P`).

1. Markdown Rendering

Bold, *italic*, ~~strikethrough~~, `inline code`

Blockquotes work too.

Feature	Status	Remark
Markdown	✓	A
Mermaid	✓	B
PlantUML	✓	C
SVG	✓	D
WaveDrom	✓	E
PDF Export	✓	F

1. Ordered list item
2. Another item
3. Third item

- Unordered item
- Another item
- Item

日本語テスト

- 日本語あいうえお
 - 2byte文字 テスト

Task Lists

- ☒ Markdown rendering
- ☒ Mermaid diagrams
- ☒ PlantUML diagrams
- ☒ WaveDrom timing diagrams

- ☒ LaTeX math
- ☒ Footnotes
- ☐ no checked

Footnotes

This sentence has a footnote^[1] and another one^[2].

Emoji

😄 🚀 👍 ⚠️ ❤️ ☆ 🔥 ☕

Math (KaTeX)

Inline math: $E = mc^2$ and $\sum_{i=1}^n i = \frac{n(n+1)}{2}$

Display math:

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Definition Lists

Markdown

A lightweight markup language for creating formatted text.

PlantUML

A tool for creating UML diagrams from plain text.

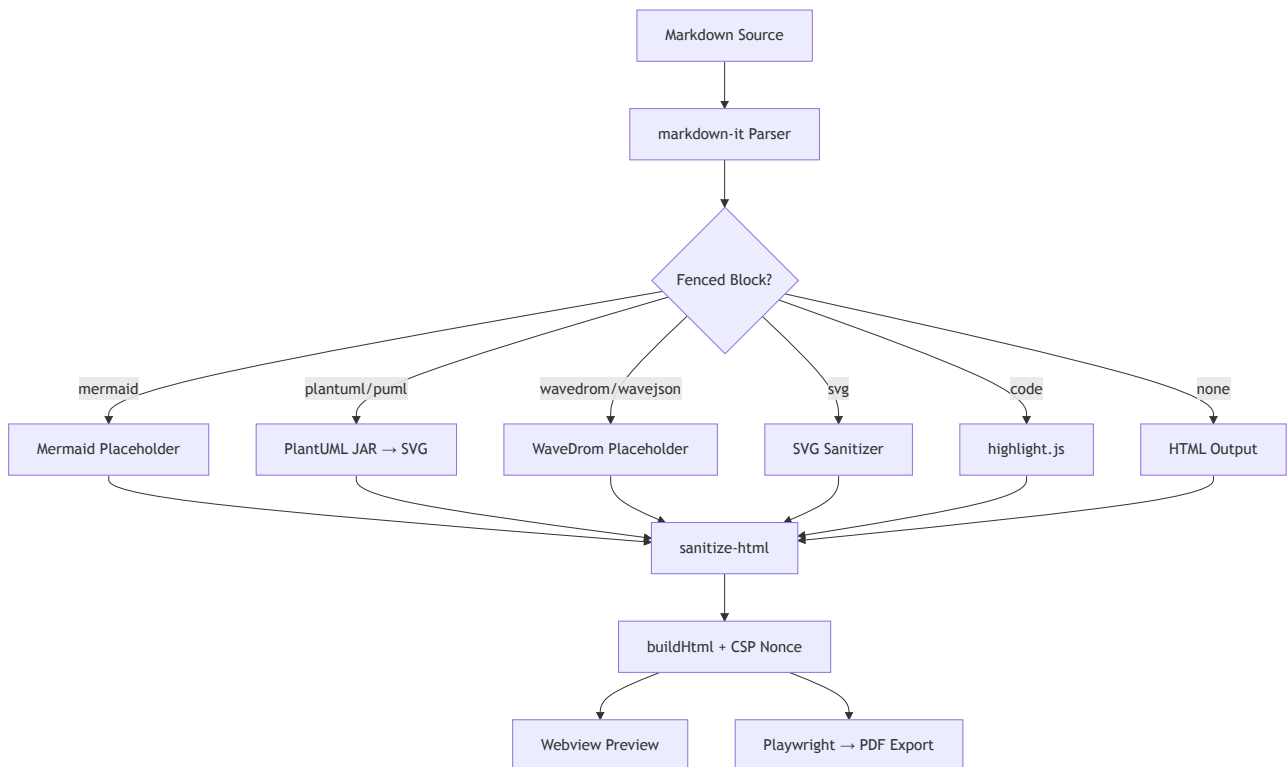
Uses Smetana layout engine (no Graphviz needed).

Superscript / Subscript

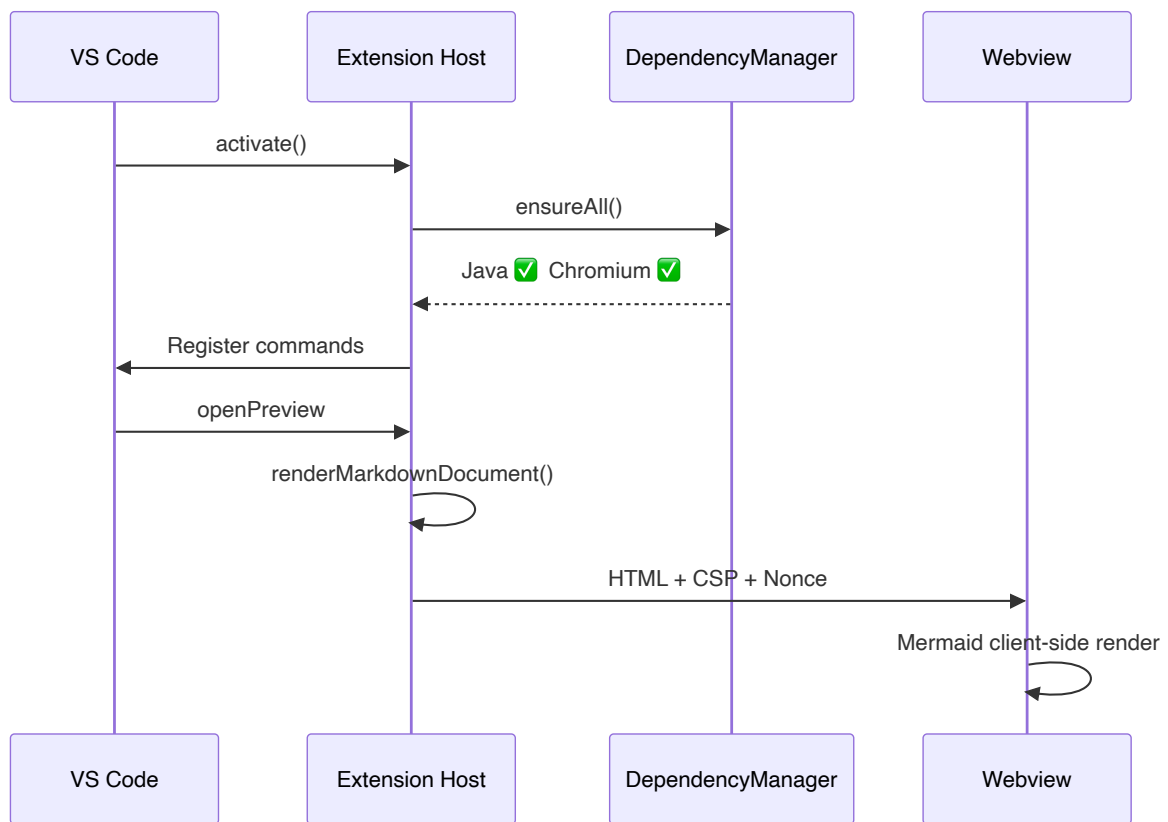
- Water: H₂O
- Einstein: E = mc²
- 19th century

2. Mermaid Diagrams

Markdown Studio Architecture



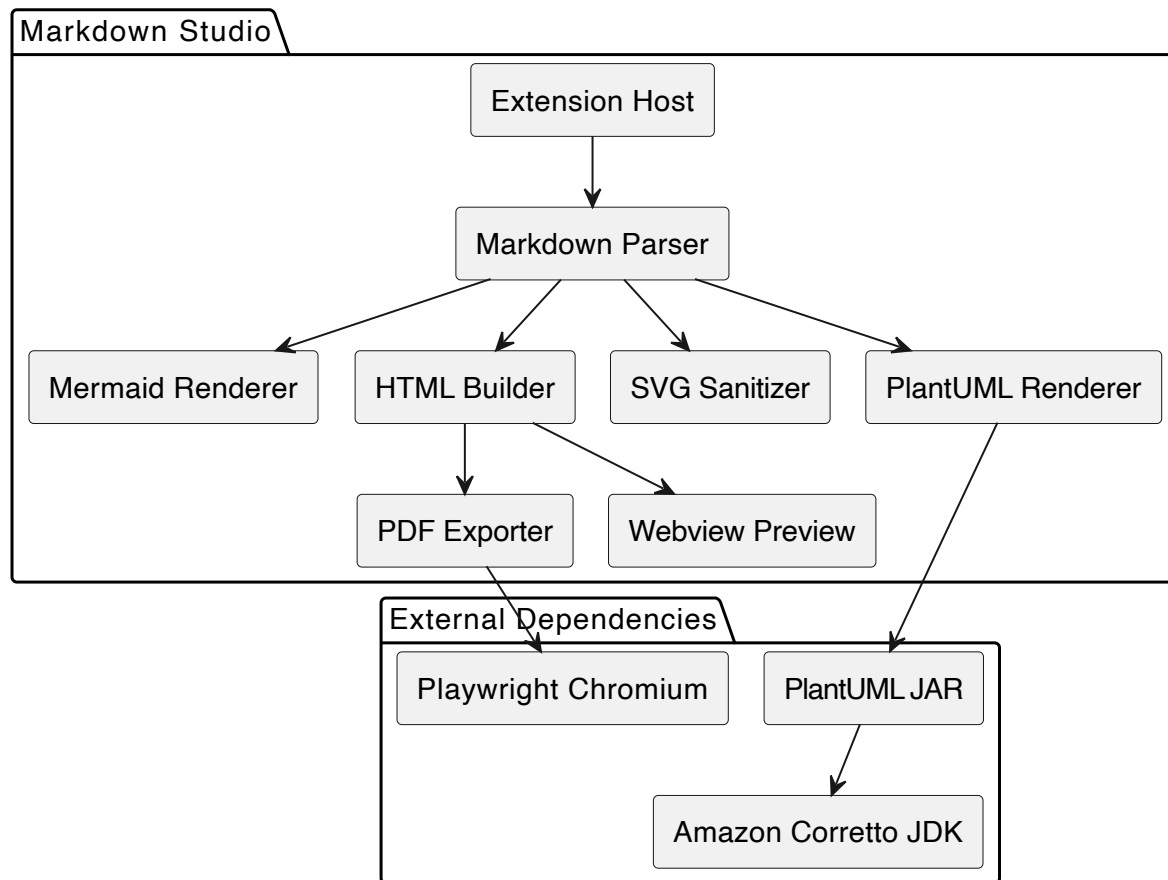
Extension Activation Flow



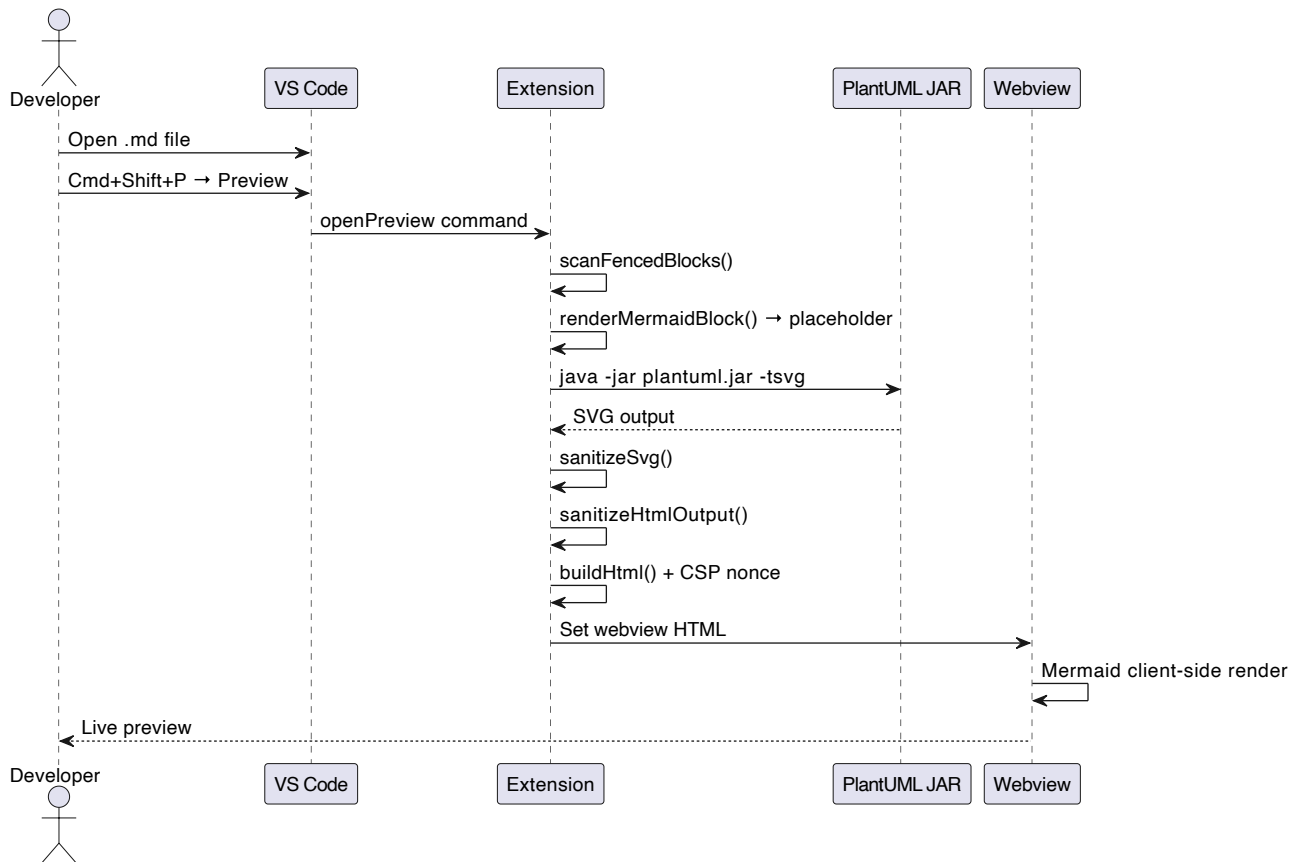
3. PlantUML Diagrams

Rendered locally via bundled JAR. No remote server.

Extension Component Diagram

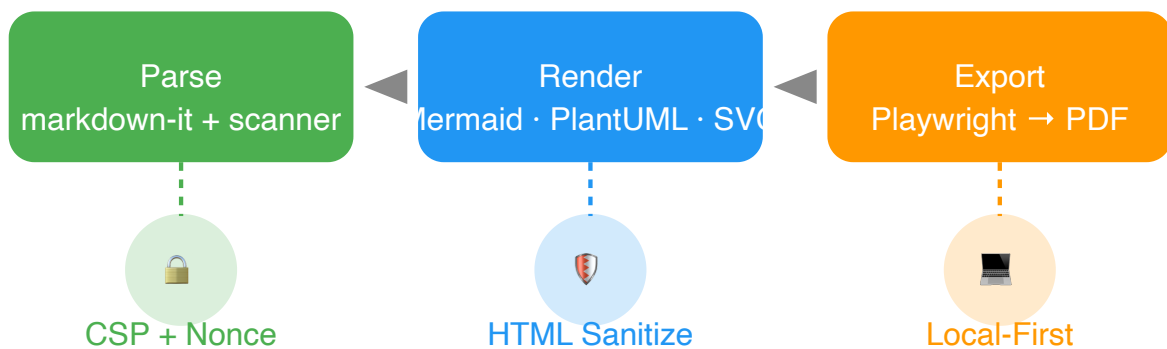


Document Processing Sequence



4. Inline SVG (Sanitized)

Dangerous elements (`<script>` , `<foreignObject>` , event handlers) are stripped automatically.



5. Syntax Highlighting

```

1 import * as vscode from 'vscode';
2
3 export function activate(context: vscode.ExtensionContext): void {
4     const depManager = new DependencyManager();
5     const status = await depManager.ensureAll(context);
6
7     context.subscriptions.push(
8         vscode.commands.registerCommand('markdownStudio.openPreview', async () => {
9             await openPreviewCommand(context);
10        }),
11        vscode.commands.registerCommand('markdownStudio.exportPdf', async () => {
12            await exportPdfCommand(context);
13        })
14    );
15 }

```

```

1 {
2     "markdownStudio.plantuml.mode": "bundled-jar",
3     "markdownStudio.java.path": "java",
4     "markdownStudio.export.pageFormat": "A4",
5     "markdownStudio.security.externalResources.mode": "whitelist"
6 }

```

```

1 # PlantUML rendering is also useful for Python projects
2 from dataclasses import dataclass
3
4 @dataclass
5 class DiagramConfig:
6     mode: str = "bundled-jar"
7     java_path: str = "java"
8     timeout_ms: int = 15000

```

```

1 fmt.Println("Hello, World!")

```

```

1 plain text without language

```

```

1 fn main() {
2     println!("Hello from Rust");
3 }

```

```

1 SELECT id, name FROM users WHERE active = true;

```

```

1 FROM node:20-alpine
2 WORKDIR /app
3 COPY . .
4 RUN npm install

```


6. Images and Security

Markdown Studio is local-first. The default security policy (`externalResources.mode = "whitelist"`) allows local files and selected trusted remote domains.

✅ Local image (relative path — works)

- local.svg

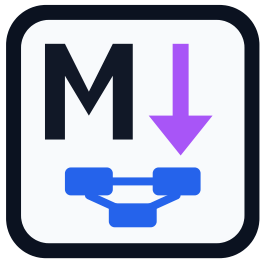


Markdown
Studio
Local-First

- icon.png



- icon.svg



Relative paths from the Markdown file are resolved and displayed correctly.

✅ Local image (absolute path — works)

Images with absolute file paths also work in the preview.

❌ External image (blocked by policy)

External image blocked by policy.

External URLs outside the allowlist are blocked by default. The image above shows a policy notice instead.

❌ External link (blocked by policy)

External link — blocked by default 🚫

External links outside the allowlist are replaced with a blocked notice in the preview.

How to allow external resources

Set `markdownStudio.security.externalResources.mode` to `allow-all` in VS Code settings to allow external images and links. This is not recommended for corporate/secure environments.

Security Summary

Resource Type	Default	Configurable
Local images (relative path)	✅ Allowed	—
Local images (absolute path)	✅ Allowed	—
External images (https://)	❌ Blocked unless allowlisted	<code>externalResources.mode: allow-all</code>
External links (https://)	❌ Blocked unless allowlisted	<code>externalResources.mode: allow-all</code>
Inline SVG	✅ Allowed	—
Mermaid diagrams	✅ Allowed	—
PlantUML diagrams	✅ Allowed	—
WaveDrom diagrams	✅ Allowed	—

7. Theme Adaptability

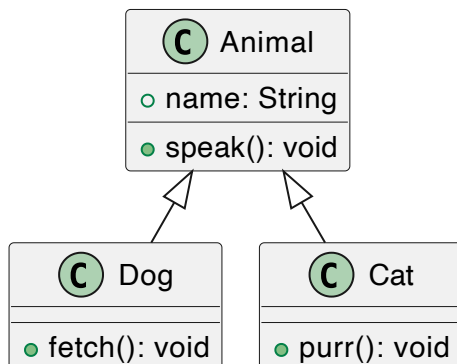
Switch between **light** and **dark** mode in VS Code (`Cmd+K Cmd+T`) to see how the preview adapts:

- Mermaid diagrams automatically switch between light and dark themes
- WaveDrom diagrams keep a light timing-diagram surface for readability
- SVG elements use colors chosen for visibility in both themes
- Code blocks use theme-aware syntax highlighting
- PlantUML output receives CSS overrides for dark mode

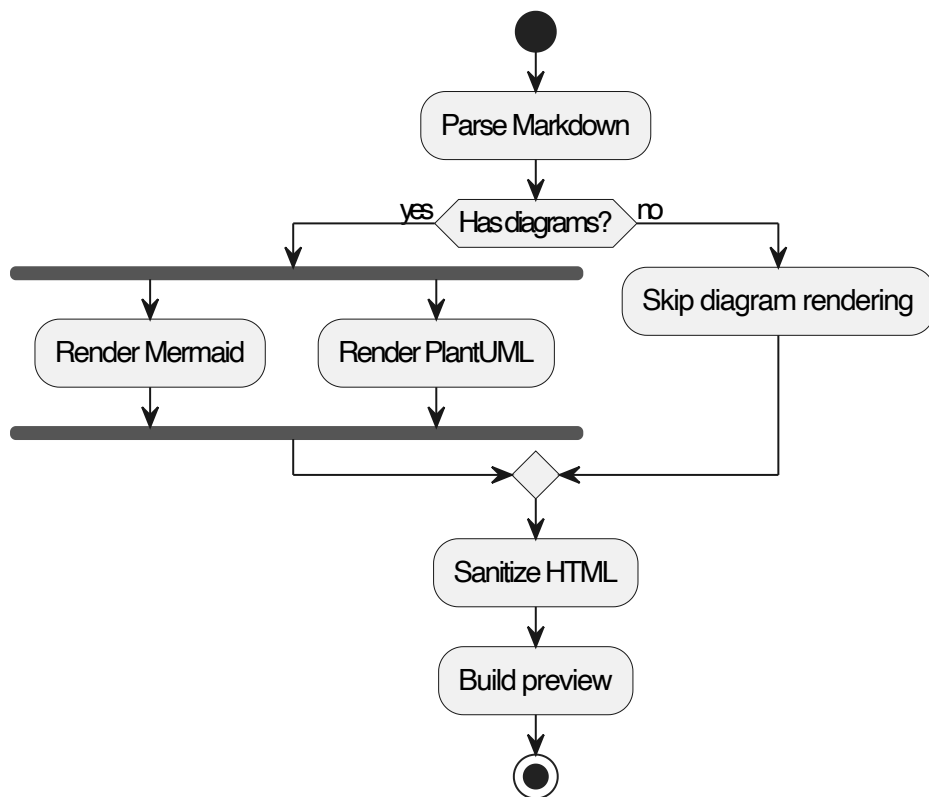
8. Diagram Type Catalog

All diagram types below are verified to render locally. PlantUML uses the bundled Smetana engine, Mermaid and WaveDrom render in the webview runtime, and inline SVG is passed through directly.

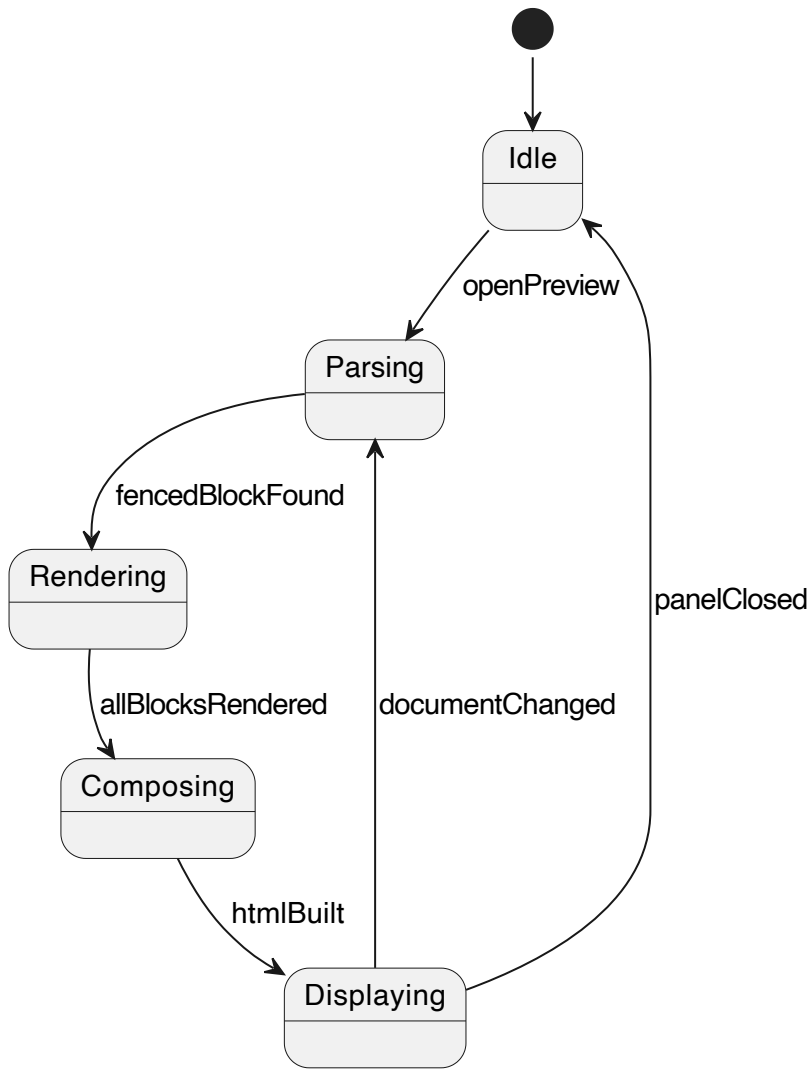
▼ Class Diagram



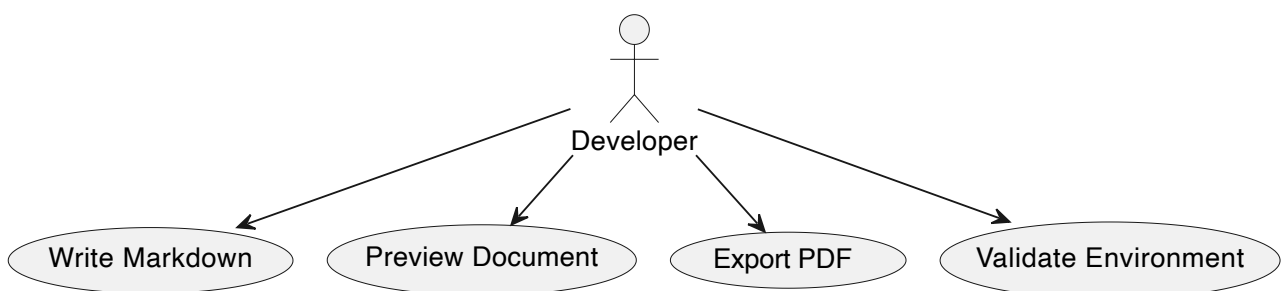
▼ Activity Diagram



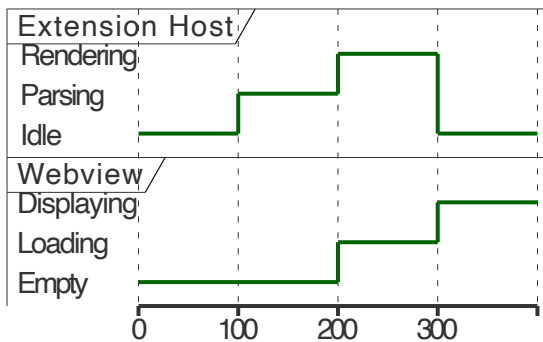
▼ State Diagram



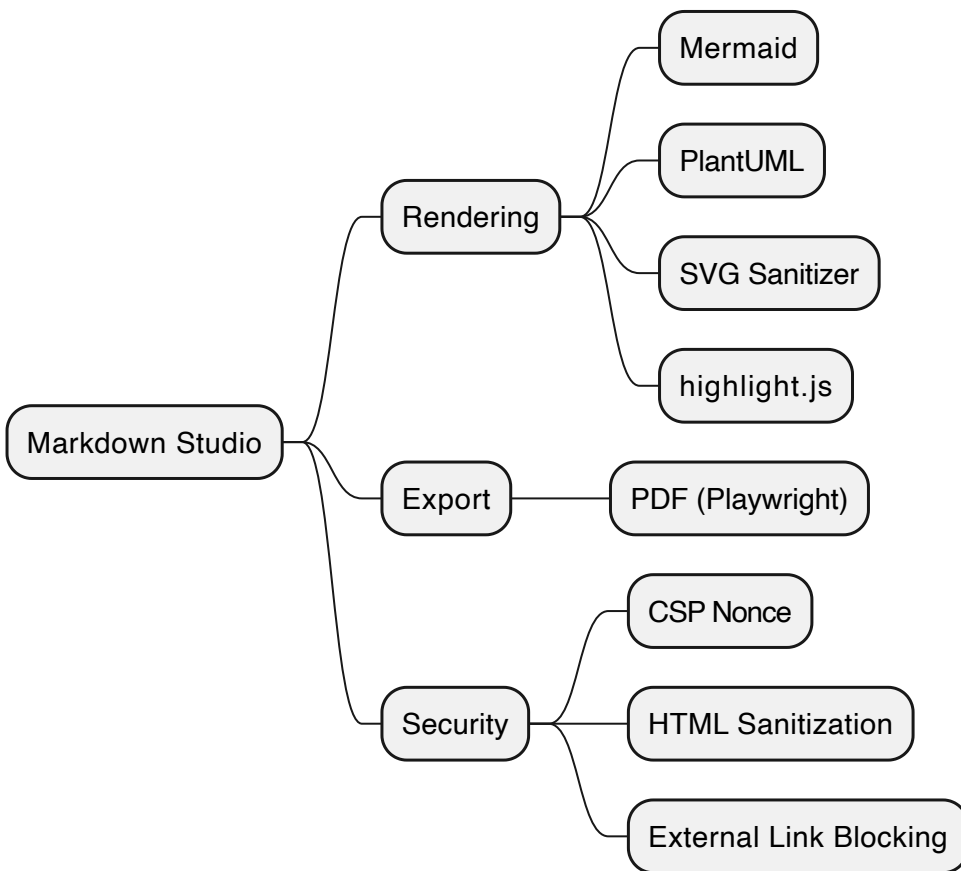
▼ Use Case Diagram



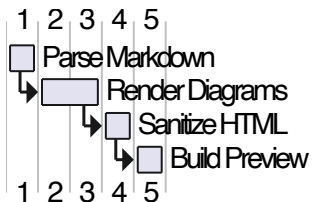
▼ Timing Diagram



▼ Mind Map



▼ Gantt Chart



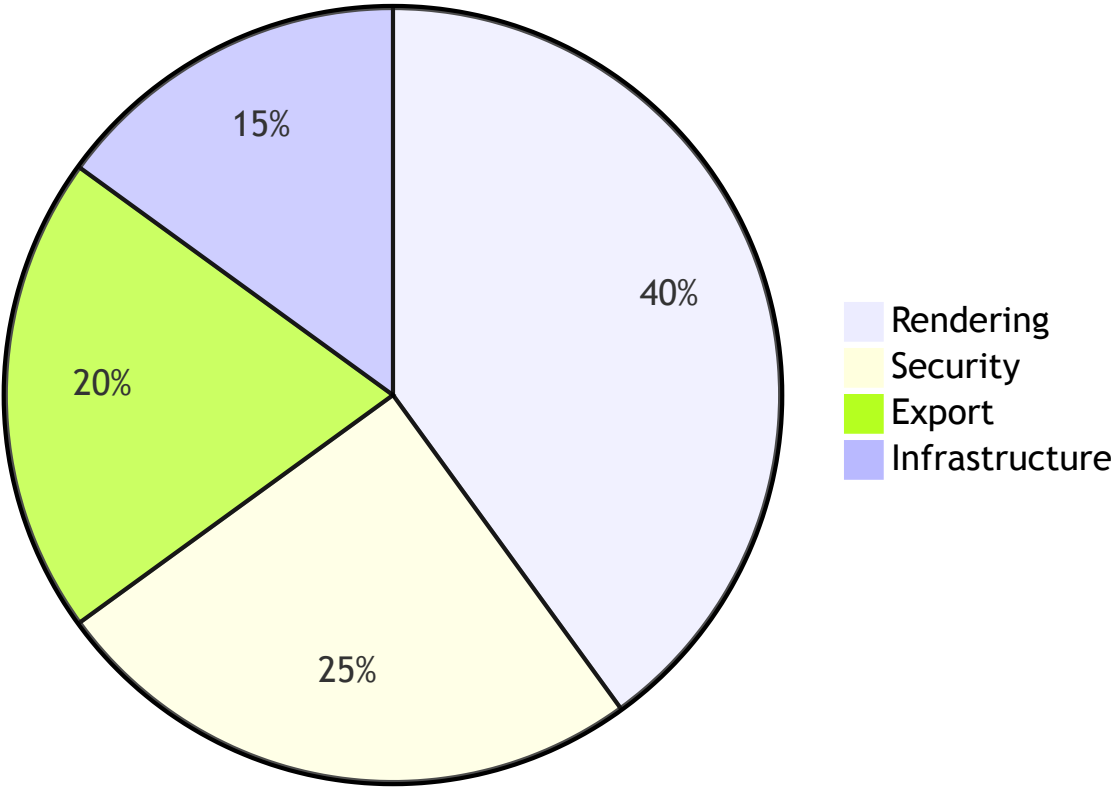
▼ Object Diagram

ExtensionConfig
plantUmlMode = "bundled-jar"
javaPath = "java"
pageFormat = "A4"
blockExternalLinks = true

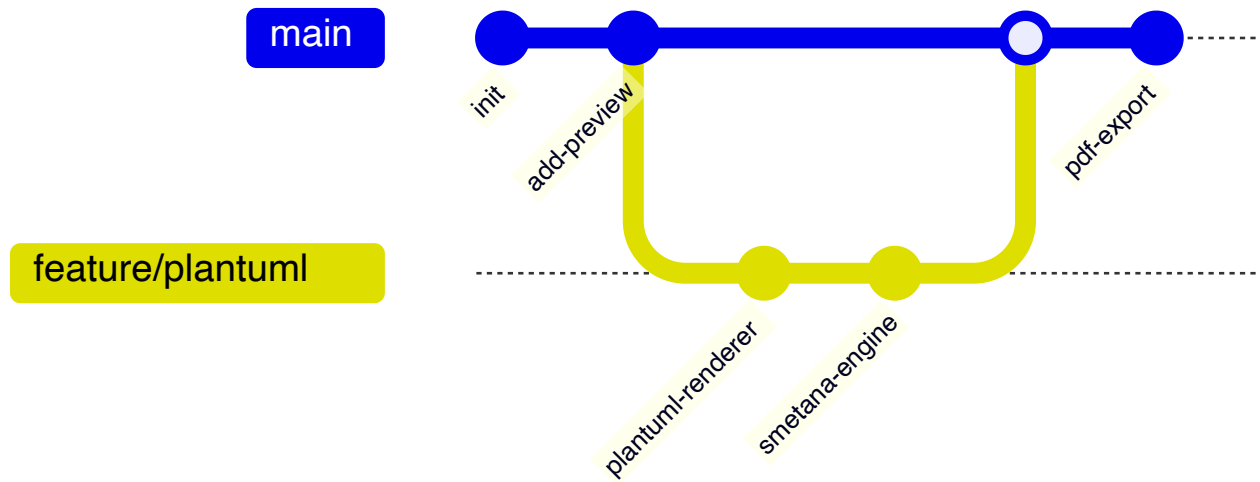
DependencyStatus
allReady = true
javaPath = "/path/to/java"
browserPath = "/path/to/chromium"

▼ Mermaid: Pie Chart

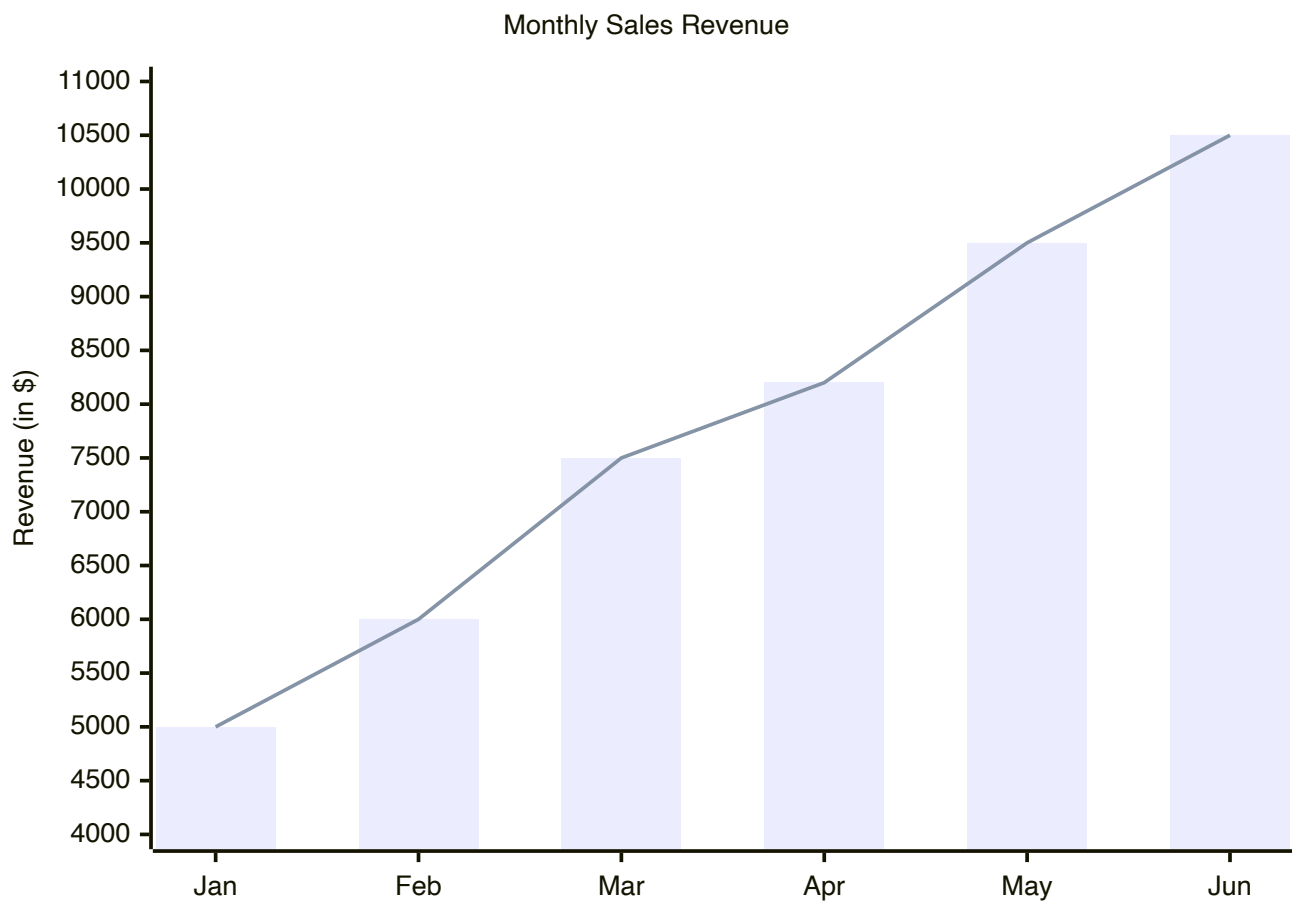
Markdown Studio Components



▼ Mermaid: Git Graph



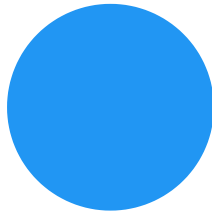
▼ Mermaid: XY Chart



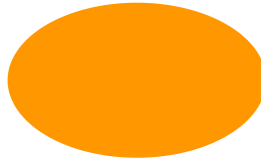
▼ SVG: Basic Shapes



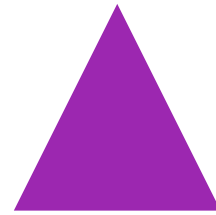
rect



circle

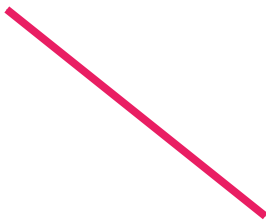


ellipse

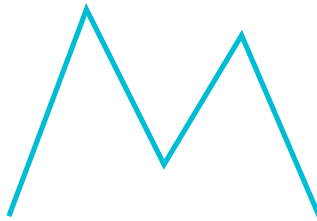


polygon

▼ SVG: Lines and Paths



line



polyline



path (curve)

▼ SVG: Text and Styling

Bold Title

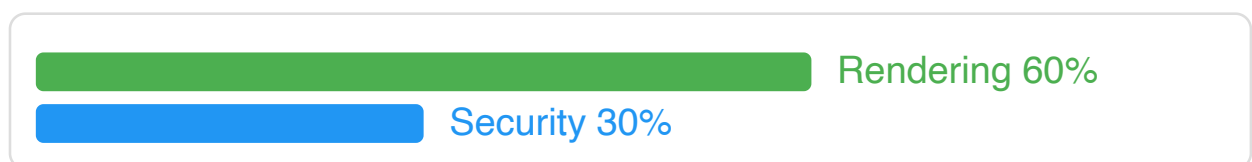
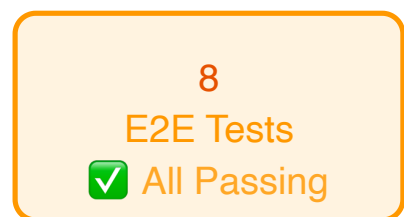
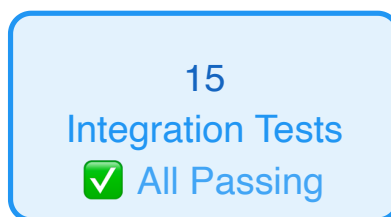
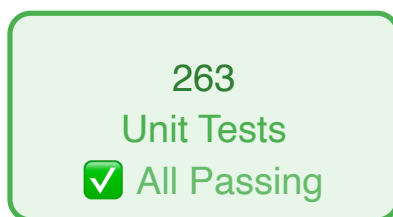
Regular body text with **colored** spans

monospace: `const x = 42;`



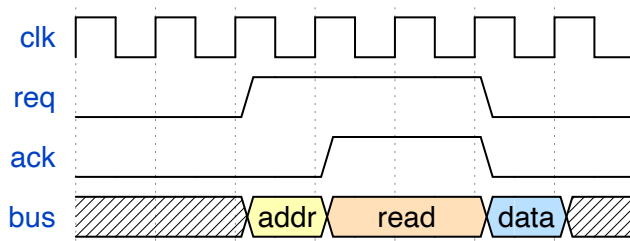
dashed border

▼ SVG: Dashboard Layout

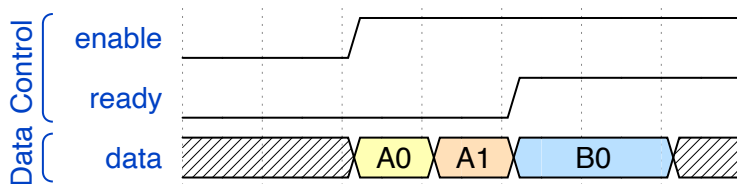


WaveDrom Timing Diagram

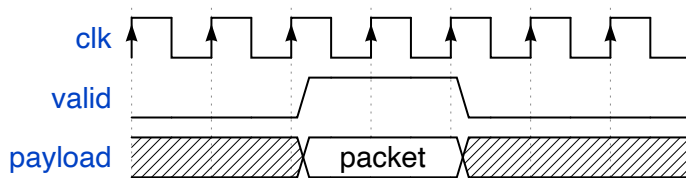
Basic Handshake



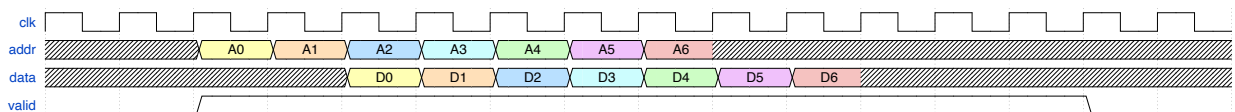
Grouped Signals via wavejson Alias



Strict JSON via wavedrom-json Alias



Wide WaveDrom Bus



Wide Diagrams for PDF Aspect Ratio

These examples are intentionally wider than the printable page. They verify that PDF export scales diagrams down without squashing their aspect ratio.

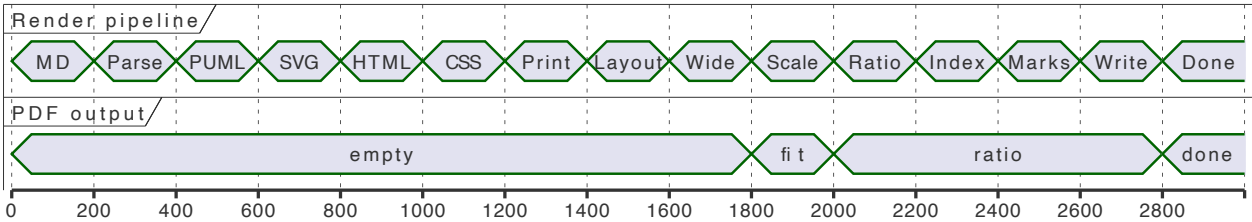
▼ Wide Mermaid flowchart



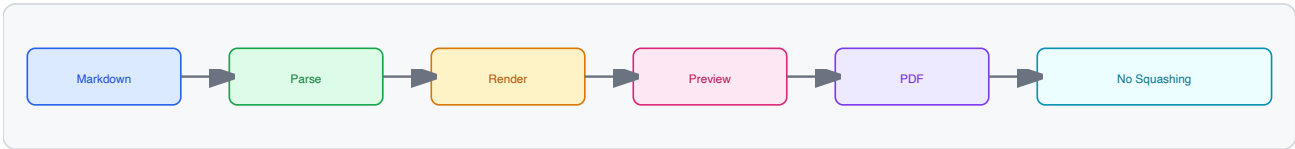
▼ Wide PlantUML component chain



▼ Wide PlantUML timing chart



▼ Wide SVG with fixed inline dimensions



9. Custom CSS

Markdown Studio supports built-in themes and inline CSS customization.

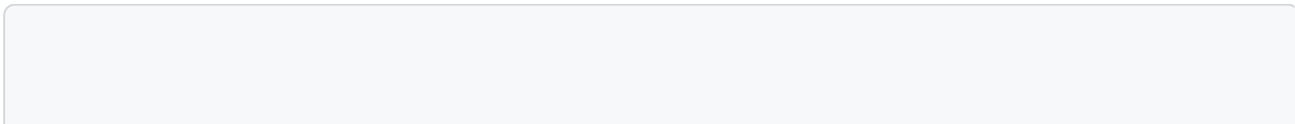
Bundled Themes

Select a theme from the settings dropdown (`markdownStudio.style.theme`):

Theme	Description
default	No extra styling — uses the selected preset only
modern	Indigo accents, soft shadows, refined typography
markdown-pdf	Classic look matching the popular Markdown PDF extension
minimal	Bare-bones, clean starting point

All themes include dark mode support and print-optimized styles.

Setup



```
1 // Pick a theme from the dropdown
2 "markdownStudio.style.theme": "modern"
3
4 // Optionally add your own CSS on top
5 "markdownStudio.style.customCss": "h1 { color: navy; } blockquote { border-left: 3px solid orange; }"
```

How It Works

- Theme CSS is injected **after** the preset styles, so theme rules take priority
- `customCss` is applied **after** the theme, so your inline rules override everything
- The same CSS applies to both preview and PDF export for consistent output
- Copy any bundled theme and customize it to create your own

10. PDF Export

This entire document can be exported to PDF:

1. Open this file in VS Code
2. `Cmd+Shift+P` → **Markdown Studio: Export PDF**
3. A `demo.pdf` will be generated next to this file

The PDF uses the same HTML pipeline as the preview.

Markdown Studio v0.8.0

1. This is the first footnote content. ↗
2. This is the second footnote with **bold** text. ↗